



HANDOUT

ELECTRO MAGNETIC FORCE & MOVE ON CHARGE OR CURRENT (EMF)

1. MAGNETIC CLASSIFICATION OF SUBSTANCES HYSTERESIS:

(A) Classification of substances according to their magnetic behaviour:

All substance show magnetic properties. An iron nail brought near a pole of a bar magnet is strongly attracted by it and sticks to it, Similar is the behaviour of steel, cobalt and nickel. Such substance are called 'ferromagnetic' substance. Some substances are only weakly attracted by a magnet, while some are repelled by it. They are called 'paramagnetic' and 'diamagnetic' substance respectively. All substance, solids, liquids and gases, fall into one or other of these classes.

- (i) **Diamagnetic substance** : Some substance, when placed in a magnetic field, are feebly magnetised opposite to the direction of the magnetising field. These substances when brought close to a pole of a powerful magnet, are somewhat repelled away from the magnet. They are called 'diamagnetic' substances and their magnetism is called the 'diamagnetism'.

Examples of diamagnetic substances are bisuth, zinc, copper, silver, gold, lead, water, mercury, sodium chloride, nitrogen, hydrogen, etc.

- (ii) **Paramagnetic substances** : Some substance when placed in a magnetic field, are feebly magnetised in the direction of the magnetising field. These substance, when brought close to a pole of a powerful magnet, are attracted towards the magnet. These are called 'paramagnetic' substance and their magnetism is called 'paramagnetism'.

- (iii) **Ferromagnetic substances** : Some substance, when placed in a magnetic field, are strongly magnetised in the direction of the magnetising field. They are attracted fast towards a magnet when brought close to either of the poles of the magnet. These are called 'ferromagnetic' substances and their magnetism is called 'ferromagnetism'.

(B) Some important terms used in magnetism :

MAGNETISATION AND MAGNETIC INTENSITY

The earth abounds with a bewildering variety of elements and compounds. In addition, we have been synthesising new alloys, compounds and even elements. One would like to classify the magnetic properties of these substances. In the present section, we define and explain certain terms which will help us to carry out this exercise.

We have seen that a circulating electron in an atom has a magnetic moment. In a bulk material, these moments add up vectorially and they can give a net magnetic moment which is non-zero.

Magnetisation

Magnetisation M of a sample to be equal to its net magnetic moment per unit volume:

$$M = \frac{m_{\text{net}}}{V}$$

M is a vector with dimensions $L^{-1} A$ and is measured in a units of $A m^{-1}$. Consider a long solenoid of n turns per unit length and carrying a current I . The magnetic field in the interior of the solenoid was shown to be given by $B_0 = \mu_0 nI$

If the interior of the solenoid is filled with a material with non-zero magnetisation, the field inside the solenoid will be greater than B_0 . The net B field in the interior of the solenoid may be expressed as

$$B = B_0 + B_m$$

where B_m is the field contributed by the material core. It turns out that this additional field B_m is proportional to the magnetisation M of the material and is expressed as $B_m = \mu_0 M$

where μ_0 is the same constant (permeability of vacuum) that appears in Biot-Savart's law.



Magnetic Intensity

It is convenient to introduce another vector field H , called the magnetic intensity, which is defined by

$$\Rightarrow H = \frac{B}{\mu} \quad \Rightarrow \quad H = \frac{B_0 + B_m}{\mu}$$

where H has the same dimensions as M and is measured in units of $A\ m^{-1}$. Thus, the total magnetic field B is written as : $B = B_0 + B_m$

We repeat our defining procedure. We have partitioned the contribution to the total magnetic field inside the sample into two parts: one, due to external factors such as the current in the solenoid. This is represented by H . The other is due to the specific nature of the magnetic material, namely M .

$$\Rightarrow H = \frac{\mu_0 H + \mu_0 M}{\mu} \Rightarrow H = \frac{H}{\mu_r} + \frac{M}{\mu_r} \Rightarrow (\mu_r - 1)H = M \Rightarrow M = \chi H$$

The latter quantity can be influenced by external factors. This influence is mathematically expressed as $M = \chi H$

where χ , a dimensionless quantity, is appropriately called the **magnetic susceptibility**. It is a measure of how a magnetic material responds to an external field. Table lists χ for some elements. It is small and positive for materials, which are called paramagnetic. It is small and negative for materials, which are termed diamagnetic.

$$\mu_r - 1 = \chi, \quad \mu_r = 1 + \chi$$

is a dimensionless quantity called the relative magnetic permeability of the substance. It is the analog of the dielectric constant in electrostatics. The magnetic permeability of the substance is μ and it has the same dimensions and units as μ_0 ; $\mu = \mu_0 \mu_r = \mu_0 (1 + \chi)$.

$$\mu_r = (1 + \chi).$$

The three quantities χ , μ_r and μ are interrelated and only one of them is independent. Given one, the other two may be easily determined.

Table :1 Magnetic Susceptibility of Some Element At 300K

Diamagnetic substance	χ	Paramagnetic substance	χ
Bismuth	-1.66×10^{-5}	Aluminium	2.3×10^{-5}
Copper	-9.8×10^{-6}	Calcium	1.9×10^{-5}
Diamond	-2.2×10^{-5}	Chromium	2.7×10^{-4}
Gold	-3.6×10^{-5}	Lithium	2.1×10^{-5}
Lead	-1.7×10^{-5}	Magnesium	1.2×10^{-5}
Mercury	-2.9×10^{-5}	Niobium	2.6×10^{-5}
Nitrogen (STP)	-5.0×10^{-9}	Oxygen (STP)	2.1×10^{-6}
Silver	-2.6×10^{-5}	Platinum	2.9×10^{-4}
Silicon	-4.2×10^{-6}	Tungsten	6.8×10^{-5}

Example 1. A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate

- (a) H (b) M (c) B
(d) the magnetising current I_m .

Solution :

(a) The field H is dependent of the material of the core, and is
 $H = nI = 1000 \times 2.0 = 2 \times 10^3\ A/m$.

(b) The magnetic field B is given by
 $B = \mu_r \mu_0 H$
 $= 400 \times 4\pi \times 10^{-7}\ (N/A^2) \times 2 \times 10^3\ (A/m) = 1.0\ T$



- (c) Magnetisation is given by

$$M = (B - \mu_0 H) / \mu_0$$

$$= (\mu_r \mu_0 H - \mu_0 H) / \mu_0 = (\mu_r - 1)H = 399 \times H \approx 8 \times 10^5 \text{ A/m}$$
- (d) The magnetising current I_M is the additional current that needs to be passed through the windings of the solenoid in the absence of the core which would give a B value as in the presence of the core. Thus

$$B = \mu_r \mu_0 (I + I_M).$$
 Using $I = 2 \text{ A}$, $B = 1 \text{ T}$, we get $I_M = 794 \text{ A}$.

MAGNETIC PROPERTIES OF MATERIALS

The discussion in the previous section helps us to classify materials as diamagnetic, paramagnetic or ferromagnetic. In terms of the susceptibility χ , a material is diamagnetic if χ is negative, para- if χ is positive and small, and ferro- if χ is large and positive.

A glance at Table 5.3 gives one a better feeling for these materials. Here ε is a small positive number introduced to quantify paramagnetic materials. Next, we describe these materials in some detail.

Table :2

Diamagnetic	Paramagnetic	Ferromagnetic
$-1 \leq \chi < 0$	$0 < \chi < \varepsilon$	$\chi \gg 1$
$0 \leq \mu_r < 1$	$1 < \mu_r < 1 + \varepsilon$	$\mu_r \gg 1$
$\mu < \mu_0$	$\mu > \mu_0$	$\mu \gg \mu_0$

Diamagnetism :

Diamagnetic substances are those which have tendency to move from stronger to the weaker part of the external magnetic field. In other words, unlike the way a magnet attracts metals like iron, it would repel a diamagnetic substance.

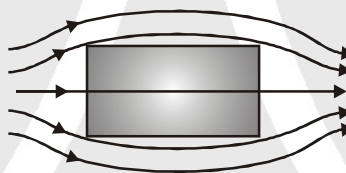
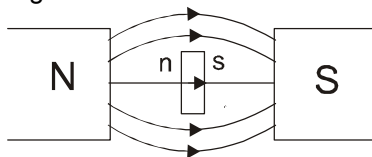


Figure shows a bar of diamagnetic material placed in an external magnetic field. The field lines are repelled or expelled and the field inside the material is reduced. In most cases, as is evident from Table 1, this reduction is slight, being one part in 10^5 . When placed in a non-uniform magnetic field, the bar will tend to move from high to low field. The simplest explanation for diamagnetism is as follows. Electrons in an atom orbiting around nucleus possess orbital angular momentum. These orbiting electrons are equivalent to current-carrying loop and thus possess orbital magnetic moment. Diamagnetic substances are the ones in which resultant magnetic moment in an atom is zero. When magnetic field is applied, those electrons having orbital magnetic moment in the same direction slow down and those in the opposite direction speed up. This happens due to induced current in accordance with Lenz's law which you will study in Electro magnetic Induction. Thus, the substance develops a net magnetic moment in direction opposite to that of the applied field and hence repulsion. Some diamagnetic materials are bismuth, copper, lead, silicon, nitrogen (at STP), water and sodium chloride. Diamagnetism is present in all the substances. However, the effect is so weak in most cases that it gets shifted by other effects like paramagnetism, ferromagnetism, etc. The most exotic diamagnetic materials are superconductors. These are metals, cooled to very low temperatures which exhibit both perfect conductivity and perfect diamagnetism. Here the field lines are completely expelled! $\chi = -1$ and $\mu_r = 0$. A superconductor repels a magnet and (by Newton's third law) is repelled by the magnet. The phenomenon of perfect diamagnetism in superconductors is called the Meissner effect, after the name of its discoverer. Superconducting magnets can be gainfully exploited in variety of situations, for example, for running magnetically levitated superfast trains.

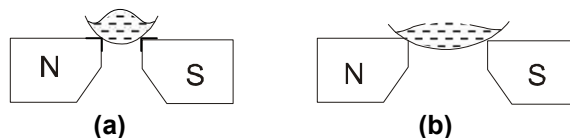
**Properties of diamagnetic substance :**

Diamagnetic substance show following properties.

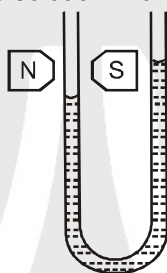
- (i) When a rod of diamagnetic material is suspended freely between two magnetic poles, then its axis becomes perpendicular to the magnetic field.



- (ii) In a non-uniform magnetic field a diamagnetic substance tends to move from the stronger to the weaker part of the field.



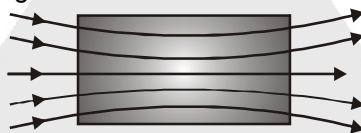
- (iii) If a diamagnetic solution is poured into a U-tube and one arm of this U-tube is placed between the poles of a strong magnet, the level of the solution in that arm is depressed.



- (iv) A diamagnetic gas when allowed to ascend in between the poles of a magnet spreads across the field.
(v) The susceptibility of a diamagnetic substance is independent of temperature.

Paramagnetism

Paramagnetic substances are those which get weakly magnetised when placed in an external magnetic field. They have tendency to move from a region of weak magnetic field to strong magnetic field, i.e., they get weakly attracted to a magnet.



The individual atoms (or ions or molecules) of a paramagnetic material possess a permanent magnetic dipole moment of their own. On account of the ceaseless random thermal motion of the atoms, no net magnetisation is seen. In the presence of an external field B_0 , which is strong enough, and at low temperatures, the individual atomic dipole moment can be made to align and point in the same direction as B_0 . Figure shows a bar of paramagnetic material placed in an external field. The field lines get concentrated inside the material, and the field inside is enhanced. In most cases, as is evident from Table 1, this enhancement is slight, being one part in 10^5 . When placed in a non-uniform magnetic field, the bar will tend to move from weak field to strong. Some paramagnetic materials are aluminium, sodium, calcium, oxygen (at STP) and copper chloride. Experimentally, one finds that the magnetisation of a paramagnetic material is inversely proportional to the absolute temperature T ,

$$M = C \frac{B_0}{T}$$

or equivalently, using Eqs $M = \chi H$ and $B_0 = \mu_0 H$

$$\chi = \frac{\mu_0}{T}$$

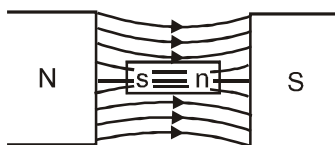
This is known as Curie's law, after its discoverer Pieree Curie (1859-1906). The constant C is called Curie's constant. Thus, for a paramagnetic material both χ and μ_r depend not only on the material, but



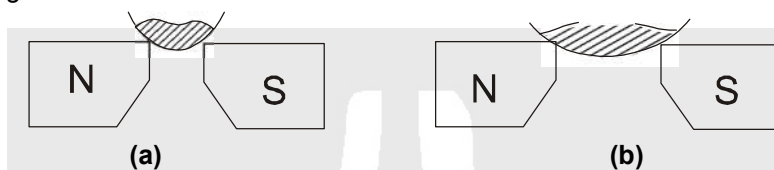
also (in a simple fashion) on the sample temperature. As the field is increased or the temperature is lowered, the magnetisation increases until it reaches the saturation value M_s , at which point all the dipoles are perfectly aligned with the field. Beyond this, Curie's law is no longer valid.

Properties of Paramagnetic Substance

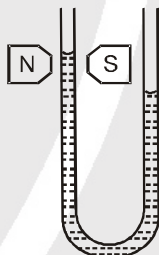
- (i) When a rod of paramagnetic material is suspended freely between two magnetic poles, then its axis becomes parallel to the magnetic field. The poles produced at the ends of the rod are opposite to the nearer magnetic poles.



- (ii) In a non-uniform magnetic field, the paramagnetic substances tend to move from weaker to stronger part of the magnetic field.



- (iii) If a paramagnetic solution is poured in a U-tube and one arm of the U-tube is placed between two strong poles, the level of the solution in that arm rises.

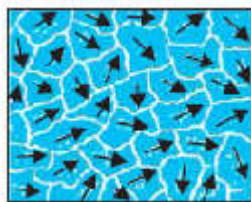


- (iv) A paramagnetic gas when allowed to ascend between the pole-pieces of a magnet spreads along the field.
- (v) The susceptibility of a paramagnetic substance varies inversely as the kelvin temperature of the substance, that is,

$$\chi_m \propto \frac{1}{T} \text{ . This known as curie's law.}$$

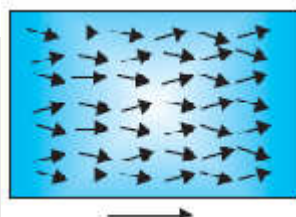
Ferromagnetism

Ferromagnetic substances are those which gets strongly magnetised when placed in an external magnetic field. They have strong tendency to move from a region of weak magnetic field to strong magnetic field, i.e., they get strongly attracted to a magnet. The individual atoms (or ions or molecules) in a ferromagnetic material possess a dipole moment as in a paramagnetic material. However, they interact with one another in such a way that they spontaneously align themselves in a common direction over a macroscopic volume called domain. The explanation of this cooperative effect requires quantum mechanics and is beyond the scope of this textbook. Each domain has a net magnetisation. Typical domain size is 1mm and the domain contains about 10^{11} atoms. In the first instant, the magnetisation varies randomly from domain to domain and there is no bulk magnetisation. This is shown in Fig.



Randomly oriented domains,

When we apply an external magnetic field B_0 , the domains orient themselves in the direction of B_0 and simultaneously the domain oriented in the direction of B_0 grow in size. This existence of domains and their motion in B_0 are not speculations. One may observe this under a microscope after sprinkling a liquid suspension of powdered ferromagnetic substance of samples. This motion of suspension can be observed. Figure (b) shows the situation when the domains have aligned and amalgamated to form a single 'giant' domain.



Aligned domains.

Thus, in a ferromagnetic material the field lines are highly concentrated. In non-uniform magnetic field, the sample tends to move towards the region of high field. We may wonder as to what happens when the external field is removed. In some ferromagnetic materials the magnetisation persists. Such materials are called hard magnetic materials or hard ferromagnets. Alnico, an alloy of iron, aluminium, nickel, cobalt and copper, is one such material. The naturally occurring lodestone is another. Such materials form permanent magnets to be used among other things as a compass needle. On the other hand, there is a class of ferromagnetic materials in which the magnetisation disappears on removal of the external field. Soft iron is one such material. Appropriately enough, such materials are called soft ferromagnetic materials. There are a number of elements, which are ferromagnetic: iron, cobalt, nickel, gadolinium, etc. The relative magnetic permeability is >1000 !

The ferromagnetic property depends on temperature. At high enough temperature, a ferromagnet becomes a paramagnet. The domain structure disintegrates with temperature. This disappearance of magnetisation with temperature is gradual. It is a phase transition reminding us of the melting of a solid crystal. The temperature of transition from ferromagnetic to paramagnetism is called the Curie temperature T_c . Table lists the Curie temperature of certain ferromagnets. The susceptibility above the

Curie temperature, i.e., in the paramagnetic phase is described by, $\chi = \frac{C}{T - T_c}$ ($T > T_c$)

Ferromagnetic substances : These substances which are strongly attracted by a magnet, show all the properties of a paramagnetic substance to a much higher degree. For example, they are strongly magnetised in relatively weak magnetising field in the same direction as the field. They have relative permeabilities of the order of hundreds and thousands. Similarly, the susceptibilities of ferromagnetic have large positive values.

Curie temperature : Ferromagnetism decreases with rise in temperature. If we heat a ferromagnetic substance, then at a definite temperature the ferromagnetic property of the substance "suddenly" disappears and the substance becomes paramagnetic. The temperature above which a ferromagnetic substance becomes paramagnetic is called the 'Curie temperature' of the substance. The Curie temperature of iron is 770°C and that of nickel is 358°C .





Example 2. A domain in ferromagnetic iron is in the form of a cube of side length $1\mu\text{m}$. Estimate the number of iron atoms in the domain and the maximum possible dipole moment and magnetisation of the domain. The molecular mass of iron is 55 g/mole and its density is 7.9 g/cm^3 . Assume that each iron atom has a dipole moment of $9.27 \times 10^{-24}\text{ A m}^2$.

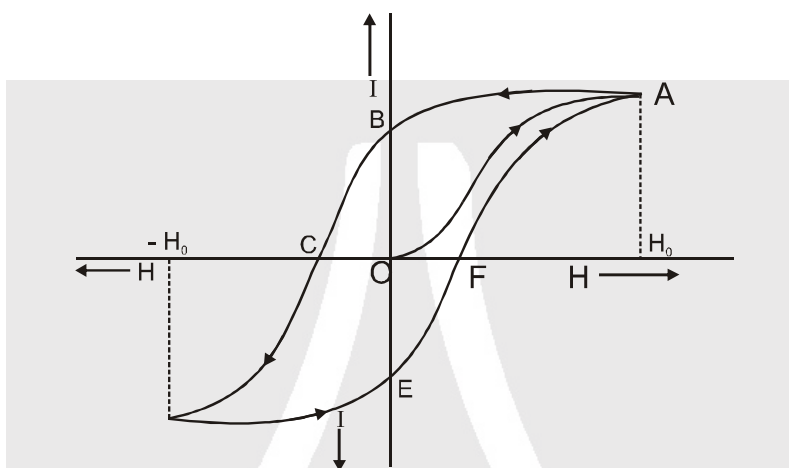
Solution : The volume of the cubic domain is $V = (10^{-6}\text{ m})^3 = 10^{-18}\text{ m}^3 = 10^{-12}\text{ cm}^3$. Its mass is volume $\times$ density $= 7.9\text{ g cm}^{-3} \times 10^{-12}\text{ cm}^3 = 7.9 \times 10^{-12}\text{ g}$. It is given that Avogadro number (6.023×10^{23}) of iron atoms have a mass of 55 g . Hence, the number of atoms in the domain is

$$N = \frac{7.9 \times 10^{-12} \times 6.023 \times 10^{23}}{55} = 8.65 \times 10^{10} \text{ atoms}$$

The maximum possible dipole moment m_{max} is achieved for the (unrealistic) case when all the atomic moments are perfectly aligned. Thus, $m_{\text{max}} = (8.65 \times 10^{10}) \times (9.27 \times 10^{-24}) = 8.0 \times 10^{-13}\text{ A m}^2$

The consequent magnetisation is $M_{\text{max}} = m_{\text{max}} / \text{Domain volume} = 8.0 \times 10^{-13}\text{ A m}^2 / 10^{-18}\text{ m}^3 = 8.0 \times 10^5\text{ A m}^{-1}$

Hysteresis



Consider that a specimen of ferromagnetic material is placed in a magnetising field, whose strength and direction can be changed. Suppose that the specimen is unmagnetised initially. When the magnetising field (H) is increased, the intensity of magnetisation (I) of the material of the specimen also increases. It is found that when the magnetising field is made zero, the intensity of magnetisation does not become zero but still has some finite value. It becomes zero only, when magnetising field is increased in reversed direction. In other words, intensity of magnetisation does not become zero on making magnetising field zero but does so a little late and this effect is called hysteresis.

The lag of intensity of magnetisation behind the magnetising field during the process of magnetisation and demagnetisation of a ferromagnetic material is called hysteresis.

Fig shows the magnetisation curve of a ferromagnetic material, when it is taken over a complete cycle of magnetisation (I) is also zero. As magnetising field is increased, intensity of magnetisation also increases along OA . Corresponding to point A , the intensity of magnetisation becomes maximum. The increase in value of the magnetising field beyond H_0 does not produce any increase in the intensity of magnetisation. In other words, corresponding to point A , the specimen of the ferromagnetic material acquires a state of magnetic saturation. If magnetising field is now decreased slowly, intensity of magnetisation decreases but not along the path AO . It decreases along the path AB . Corresponding to point B , magnetising field becomes zero but some magnetisation equal to OB is still left in the specimen. Here, OB gives the measure of retentivity of the material of the specimen.

The value of the intensity of magnetisation of a material, when the magnetising field is reduced to zero, is called retentivity of the material. It is also known as residual magnetism or remanence.



To reduce intensity of magnetisation to zero, the magnetising field has to be increased in reverse direction. As it is done so, the intensity of magnetisation decreases along BC, till it become zero corresponding to point C. Thus, to make intensity of magnetisation zero, magnetising field equal OC has to be applied in reverse direction. Here, OC gives the measure of coercivity of the material of the specimen .

The value of reverse magnetising field required so as to reduce residual magnetism to zero, is called coercivity of the material.

When the magnetising field is further increased in reverse direction, intensity of magnetisation increase along CD with the increase in magnetising field. Corresponding to point D (when the magnetising field becomes $-H_0$) it again acquires a saturation value, which is symmetrical to that corresponding to point A. If the magnetising field is decreased from $-H_0$ to zero, the intensity of magnetisation follows the path DE. Finally, when magnetising field is increased field in original direction, the point A is reached via EFA. If the magnetising field is repeatedly changed between H_0 and $-H_0$, the curve ABCDEFA is retraced. The curve ABCDEFA is called the hysteresis loop. It is found that the area of the hysteresis loop (I – H curve) is proportional to the net energy absorbed per unit volume by the specimen, as it taken over a complete cycle of magnetisation and demagnetisation. The energy so absorbed by the specimen appears as the heat energy.

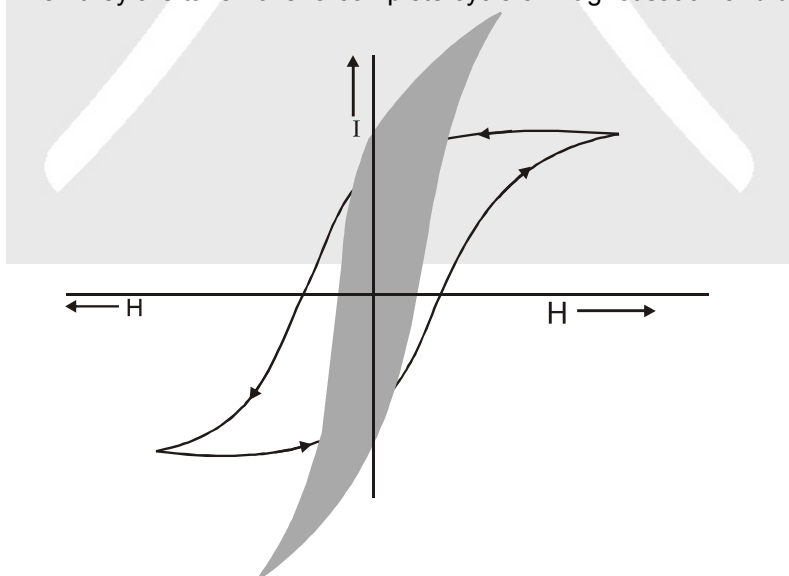
Note :- If hysteresis loop is drawn by plotting a graph between magnetic induction (B) and intensity of magnetisation, then area of the hysteresis loop is numerically equal to the work done per unit volume (or energy absorbed per unit volume) in taking the magnetic specimen over a complete cycle of magnetisation.

COMPARISON OF HYSTERESIS LOOPS FOR SOFT IRON AND STEEL :-

The shape of the hysteresis loop is a characteristics of a ferromagnetic substance. It gives the idea about many important magnetic properties of the substance.

Figure Shown hysteresis loops for soft iron and steel. Whereas the hysteresis loop for soft iron is narrow, the hysteresis loop for steel is quite wide. The following conclusion can be drawn from the study of the hysteresis loops of soft iron and steel :

1. The area of hysteresis loop for soft iron is much smaller then that for steel. Therefore, loss of energy per unit volume in case of soft iron will be very small as compared to that in case of steel, when they are taken over a complete cycle of magnetisation and demagnetisation.





2. Soft iron acquires maximum intensity of magnetisation for comparatively much lesser value of magnetising field than in case of steel. In other words, soft iron is much strongly magnetised (or more susceptible to magnetisation) than steel.
3. The retentivity of soft iron is greater than that of steel. On removing magnetising field, quite a large amount of magnetisation is retained by soft iron.
4. The coercivity of steel is much larger than that of soft iron. Therefore, the residual magnetism in steel can not be destroyed that easily as in case of soft iron.

Section of magnetic materials :

The choice of a magnetic material for making permanent magnet, electromagnet, core of transformer or diaphragm of telephone ear-piece can be decided from the hysteresis curve of the material.

(i) Permanent magnets :

The material for a permanent magnet should have high retentivity so that the magnet is strong, and high coercivity so that the magnetisation is not wiped out by stray external fields, mechanical ill treatment and temperature changes. The hysteresis loss is immaterial because the material in this case is never put to cyclic changes of magnetisation. From these considerations permanent magnets are made of steel. The fact that the retentivity of soft iron is a little greater than that of steel is outweighed by its much smaller coercivity, which makes it very easy to demagnetise.

(ii) Electromagnets :

The material for the cores of electromagnets should have high permeability (or high susceptibility), specially at low magnetising fields, and a low retentivity. Soft iron is suitable material for electromagnets).

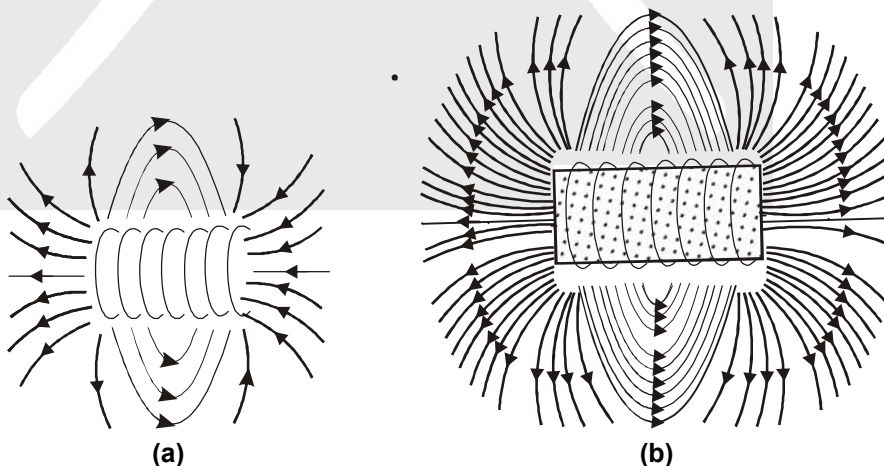
(iii) Transformer cores and telephone diaphragms :

In these cases the material goes through complete cycles of magnetisation continuously. The material must therefore have a low hysteresis loss to have less dissipation of energy and hence a small heating of the material (otherwise the insulation of windings may break), a high permeability (to obtain a large flux density at low field) and a high specific resistance (to reduce eddy current losses).

Soft-iron is used for making transformer cores and telephone diaphragms : More effective alloys have now been developed for transformer cores. They are permalloys, mumetals etc.

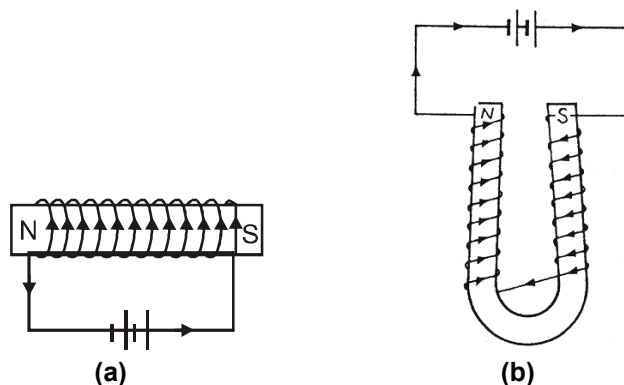
2. ELECTROMAGNET :

If we place a soft-iron rod in the solenoid, the magnetism of solenoid increases hundreds of times. Then the solenoid is called an 'electromagnet'. It is a temporary magnet.





An electromagnet is made by winding closely a number of turns of insulated copper wire over a soft-iron straight rod or a horse-shoe rod. On passing current through this solenoid, a magnetic field is produced in the space within the solenoid.



Example 3. A magnetising field of 1600 Am^{-1} produces a magnetic flux of $2.4 \times 10^{-5} \text{ wb}$ in an iron bar of cross-sectional area 0.2 cm^2 . Calculate permeability and susceptibility of the bar.

Solution :
$$B = \frac{\Phi}{A} = \frac{2.4 \times 10^{-5} \text{ Wb}}{0.2 \times 10^{-4} \text{ m}^2} = 1.2 \text{ Wb/m}^2 = 1.2 \text{ N A}^{-1} \text{ m}^{-1}.$$

The magnetising field (or magnetic intensity) H is 1600 Am^{-1} . Therefore, the magnetic permeability is given by -

$$\mu = \frac{B}{H} = \frac{1.2 \text{ N A}^{-1} \text{ m}^{-1}}{1600 \text{ Am}^{-1}} = 7.5 \times 10^{-4} \text{ N/A}^2.$$

Now, from the relation $\mu = \mu_0 (1 + \chi_m)$, the susceptibility is given by

$$\chi_m = \frac{\mu}{\mu_0} - 1.$$

We know that $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$. $\therefore \chi_m = \frac{7.5 \times 10^{-4}}{4 \times 3.14 \times 10^{-7}} - 1 = 596.$

Example 4. The core of toroid of 3000 turns has inner and outer radii of 11 cm and 12 cm respectively. A current of 0.6 A produces a magnetic field of 2.5 T in the core. Compute relative permeability of the core. ($\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$).

Solution : The magnetic field in the empty space enclosed by the windings of a toroid carrying a current i_0 is $\mu_0 n i_0$

where n is the number of turns per unit length of the toroid and μ_0 is permeability of free space.

If the space is filled by a core of some material of permeability μ , then the field is given by

$$B = \mu n i_0$$

But $\mu = \mu_0 \mu_r$, where μ_r is the relative permeability of the core material. Thus,

$$B = \mu_0 \mu_r n i_0 \quad \text{or} \quad \mu_r = \frac{B}{\mu_0 n i_0}$$

Here $B = 2.5 \text{ T}$, $i_0 = 0.7 \text{ A}$ and $n = \frac{3000}{2\pi r} \text{ m}^{-1}$, where r is the mean radius of the toroid

$$(r = \frac{11+12}{2} = 11.5 \text{ cm} = 11.5 \times 10^{-2} \text{ m}). \text{ Thus,}$$

$$\mu_r = \frac{2.5}{(4\pi \times 10^{-7}) \times (3000/2\pi \times 11.5 \times 10^{-2}) \times 0.7} = \frac{2.5 \times 11.5 \times 10^{-2}}{2 \times 10^{-7} \times 3000 \times 0.7}$$

$$\mu_r = 684.5$$